Physics-Guided Calibrated Localization of Methane Sources

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***Abstract*—We present a physics-guided method for localizing methane leaks from inexpensive sensor networks, with a statistical guarantee that the reported region contains the source 90% of the time. The core of the method is a set of *physics input images*: for every cell of a site map, closed-form formulas score how strongly the sensor readings point to a leak at that cell, how visible such a leak would be to the sensor layout, and how well the best-fitting leak rate at that cell explains the readings. Computed in milliseconds, these images are supplied to a standard neural network as additional input channels through a small zero-initialized head (0.2% added parameters), trained with ordinary labels, and calibrated with split conformal prediction. When the wind is only measured rather than known, the images are averaged over the wind sensor's stated error range, so the network also sees how fragile each cell's evidence is to wind error. On a testbed whose physics is exactly solvable, so the smallest 90%-mass region under the exact simulator posterior is computable, the method halves the median 90% region of a calibrated deep localizer from 94–98 m to 46–48 m across three network families, at the 50 m scale that U.S. EPA super-emitter notifications use to associate a leak with a facility. At a representative 10°/10% anemometer-error setting, where the evaluated measured-wind inversion baselines produce operationally uninformative calibrated regions, the method retains its guarantee at 70–73 m, with a quarter of cases below 50 m. All results are verified against exact references and reported over multiple training seeds.**

***Index Terms*—methane monitoring, conformal prediction, sensor networks, wind uncertainty, physics-guided machine learning**

I. Introduction

Methane traps roughly 80× more heat than CO_2 over its first 20 years and fades from the atmosphere within about a decade, so stopping large leaks quickly is one of the highest-leverage climate actions available today. A large share of oil-and-gas methane comes from a small number of large point sources, known as super-emitters and defined by the U.S. EPA as events of $100\,\mathrm{kg\,h^{-1}}$ or more [2]. Detection has advanced rapidly: UNEP's satellite system issued over a thousand alerts in two years [1]. The bottleneck is what happens next: connecting an alert to a specific place that a crew can inspect. EPA's Super-Emitter Program (implementation extended to January 2027) asks notifications to identify any facility within **50 m** of the estimated leak location [2], and the EU imposes analogous source-level duties [3]. We therefore adopt 50 m as an operationally motivated target: a region small enough to trigger a focused inspection.

Continuous monitoring networks, in which a small number of inexpensive sensor masts capture the plume as the wind sweeps it across the site, are where this localization can happen. Machine learning provides instant estimates, and a standard statistical wrapper called split conformal prediction [4] adds a guarantee: a region that provably contains the true source 90% of the time. The gap we address is twofold. First, the guarantee speaks only to how *often* the region contains the source, not to how *small* the region is, and there was previously no practical way to measure how much smaller a guaranteed region could have been. Second, the classical alternative of inverting the plume physics directly requires the wind to be known exactly, while every real site has only a wind *measurement*; how each approach behaves under that everyday mismatch had not been carefully quantified.

Contributions. The paper centers on a single method and the evidence for it.

- **C1: Physics input images that carry their wind-error budget.** For every candidate cell, closed-form formulas produce four images: the evidence for a leak at that cell, the fragility of that evidence across the wind sensor's stated error range, the visibility of such a leak to the sensor layout, and the fit quality of the best leak rate at that cell. A zero-initialized head (0.2% added parameters) joins the images to any base network at the logits; training is ordinary supervised learning; split conformal prediction supplies the guarantee. When the error budget is set to zero, the images provably reduce to the classical matched-filter pair, so one recipe covers exact and measured wind alike.
- **C2: A measuring stick that quantifies the method's value.** A methane testbed (CH4-T) whose physics is exactly solvable, so the exact-model reference region is computable for every scenario and any guaranteed localizer can be audited in meters. A verified-90% deep localizer delivered 94–98 m where the reference is 6–7 m; the images close half of that gap on all three network families we tested (46–48 m).
- **C3: Robustness under measured wind.** At the evaluated 10°/10% anemometer-error setting, the measured-wind inversion baselines produce operationally uninformative calibrated regions (essentially the whole site), while the method degrades gracefully, and each added image tightens the guaranteed regions further, to 70–73 m.

Every component is standard and auditable, every number below comes from one shared, verified simulator with identical splits and calibration harness, and headline comparisons carry paired-bootstrap intervals over the same 2,000 test scenarios.

Pipeline figure placeholder: one scenario flowing left to right – site sketch (masts, wind) → the four computed physics images → set network + zero-initialized head → the guaranteed 90% region.

Fig. 1. One scenario flowing through the pipeline: sensor data → the four computed physics images (Eqs. 6–9) → any set network with the zero-initialized head (Eq. 11) → the guaranteed 90% region (Eq. 14), calibrated under the deployment wind. At ε=0 the images reduce to the classical pair (Eq. 10).

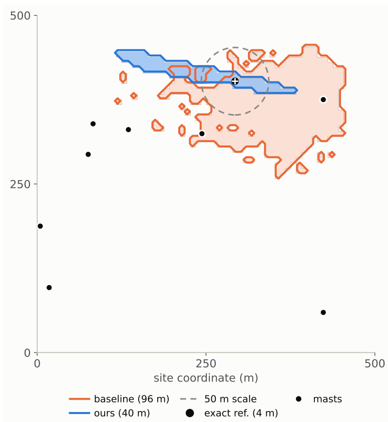

Fig. 2. One test scenario (8 masts, exact wind): 90% regions of the labels-only baseline and of our method (conformal masks), the exact-model reference (equal-area circle), and the 50 m facility scale. Selection rule: among 8-mast scenarios, the one jointly closest (log scale) to that subgroup's median baseline and method radii, i.e., a typical case, not a best case.

II. Related Work

Conformal prediction gives distribution-free coverage guarantees [4]; how small the guaranteed sets are is the open half of the problem, and our audit scores it directly against the exact probability map at matched, verified coverage. **Simulation-based inference** trains neural networks to approximate probability maps from simulated data [6], [7]; its standard diagnostics check calibration [5], and recent work examines what happens when the simulator's settings differ from reality at deployment [8]. Our measured-wind design addresses exactly that setting by building the wind sensor's declared error range into the network's inputs. Our maps are hand-derived statistics, complementing learned summaries. **Source-term estimation** inverts dispersion physics with optimization or sampling at minutes to hours per event [9]; learned localizers answer in milliseconds, and our audit adds the missing instrument for judging how much information they extract. **Controlled-release testing** (METEC) evaluates continuous monitoring systems in the field [10]; our testbed mirrors that task structure with exact ground truth, complementing rather than replacing field validation.

III. Methodology

Our methodology has three layers, and each exists for a reason. A simulation system makes the truth knowable, so that every claim in this paper can be measured rather than assumed. Machine-learning localizers make inference fast and generalizable, answering in milliseconds on whatever sensor layout a site has. Between the two we insert a physics input processor, which computes the physical relationships the network would otherwise have to rediscover from data and presents them as inputs. Physics contributes exact understanding; learning contributes calibration and generalization; the sections below construct each layer in turn and state why it is built the way it is.

A. The simulation system: making the truth knowable (C2)

CH4-T places one source of unknown grid cell $c^* \in \{1,\dots,64^2\}$ and rate q (10–500 kg h^{-1}, log-uniform, spanning the 100 kg h^{-1} super-emitter threshold) on a 500×500 m site watched by $S \in \{4,\dots,12\}$ masts over T=30 one-minute steps. Transport follows the standard Gaussian plume with published Pasquill–Gifford/Briggs dispersion (verified against the original tables; mass conservation checked to 4×10^{-6}). In plume-frame coordinates (downwind distance x, crosswind offset y_{cw}, sensor height z_r, release height H), the concentration per unit rate is

$$g \;=\; \frac{1}{2\pi U\sigma_y(x)\,\sigma_z(x)}\, e^{-\frac{y_{\mathrm{cw}}^2}{2\sigma_y^2}} \left(e^{-\frac{(z_r-H)^2}{2\sigma_z^2}} + e^{-\frac{(z_r+H)^2}{2\sigma_z^2}} \right), \tag{1}$$

where U is wind speed and σ_y, σ_z are the stability-dependent dispersion widths. Stacking all sensors and time steps, a leak at cell c produces the fingerprint $g_c(w) \in \mathbb{R}^{ST}$ under wind sequence w, and readings are

$$y \;=\; q\,g_{c^*}(w) \;+\; \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2 I). \tag{2}$$

Wind uncertainty is built into the data. Readings are generated under the true wind w, but the dataset records only the measurement $\hat{w}$, corrupted independently at every step t:

$$\hat{\theta}_t = \theta_t + \delta_t, \quad \hat{s}_t = s_t\, e^{\eta_t}, \quad \delta_t \sim \mathcal{N}(0, \sigma_\theta^2),\ \eta_t \sim \mathcal{N}(0, \sigma_s^2), \tag{3}$$

with $\sigma_\theta = 10^\circ$ and $\sigma_s = 0.10$, the accuracy class of inexpensive monitoring anemometers. No model, at any stage, observes the true wind.

The simulator is deliberately built so that the truth is computable. Because the model (2) is linear in q, the per-cell likelihood depends on the data only through three summary quantities,

$$a_c = g_c^\top y, \qquad b_c = \|g_c\|^2, \qquad \rho = \|y\|^2, \tag{4}$$

and the exact probability map under known wind follows by integrating the rate out of the Gaussian likelihood:

$$p(c \mid y) \;\propto\; \int_0^\infty \exp\Big(-\,\tfrac{\rho - 2qa_c + q^2 b_c}{2\sigma^2}\Big)\,\pi(q)\,dq, \tag{5}$$

evaluated by high-precision quadrature (normalization checked to $\sim 10^{-12}$). **The exact-model reference region.** For each scenario we normalize (5) over cells, using a uniform source-cell prior and the same log-uniform rate prior $\pi(q)$ on $[10, 500]\,\mathrm{kg\,h^{-1}}$ used to generate the data (peak-aware composite Gauss–Legendre quadrature over that range), and apply the identical conformal wrapper of §III-D to the exact map itself, with the same randomized tie handling and strict threshold rule: the region is the smallest set of highest-posterior cells at the calibrated mass threshold. We call this the *exact-model reference region*. It is an oracle reference under the CH4-T simulator assumptions, not a universal information-theoretic lower bound. The two reported values correspond to different coverage levels: 7.0 m at nominal 90% (empirical 0.952; the exact map is conservative on the discrete grid) and 6.2 m at the learned models' matched empirical coverage of 0.89. A blocking check requires the wrapper applied to the exact map to reproduce its own guarantee before any experiment proceeds; this check caught two real implementation bugs, including a tie-handling correction that we report for reuse.

B. Base learners: why machine learning, and which networks

A monitoring system must answer in seconds, on whatever sensor layout a site has, under the wind it actually measured, while physics inversion takes minutes to hours per event and requires the wind to be known. A network trained once on simulated scenarios answers in milliseconds and, because it learns from examples, absorbs regularities with no closed form, such as how anemometer error distorts the evidence. The readings form a *set* of tokens (mast position, time, concentration, wind step) whose size varies from site to site, so the natural base learners are permutation-invariant set networks. We evaluate three families with different inductive biases, DeepSets (pooling), a graph neural network (relational structure between masts), and a Set Transformer (attention), precisely so that the method's value is demonstrated independently of any one architecture choice.

C. The physics input processor (C1)

The plume model (1)–(2) fixes an exact relationship between any hypothesized leak and the readings it should produce, so testing a hypothesis is a correlation computation. The controlled comparisons of §IV-B show that networks trained end-to-end do not efficiently reconstruct this computation across 4,096 candidate cells (a correctly calibrated baseline delivers 94–98 m), while the computation by itself cannot judge how much to trust its own answer (calibrated alone, it returns the whole site). We therefore insert an *input processor* between the raw data and the network (Fig. 1): it evaluates the physics relationships explicitly, cell by cell, and presents the results as images aligned with the output grid, where a small convolutional head can read them spatially. This is the division of labor the paper centers on: physics supplies exact understanding of how a leak maps to readings, and the network supplies what physics lacks, calibration and generalization learned from examples. The classical matched-filter score of cell c normalizes the alignment a_c by the fingerprint's energy,

$$z_c \;=\; \frac{a_c}{\sigma\sqrt{b_c}}, \tag{6}$$

and $\log b_c$ records how visible a leak at c would be to the current sensor layout. These two 64×64 images are the zero-budget input maps. To carry the wind sensor's stated error range, we draw $K{=}8$ plausible true winds from the error model (3) centered on the measurement,

$$w^{(k)} \sim \mathcal{E}_\varepsilon(\hat{w}), \qquad k = 1, \dots, K, \tag{7}$$

recompute (4) under each draw, and feed four images: the average evidence and its variability,

$$\bar{z}_c = \frac{1}{K}\sum_k z_c^{(k)}, \qquad v_c = \Big(\tfrac{1}{K}\sum_k \big(z_c^{(k)} - \bar{z}_c\big)^2\Big)^{1/2}, \tag{8}$$

the average sensitivity $\overline{\log b_c}$, and the average *fit quality*. For each draw, the best nonnegative rate at cell c is $\hat{q}_c = \max(0, a_c/b_c)$, and the residual error it leaves is available in closed form,

$$R_c^{(k)} \;=\; \rho \;-\; 2\hat{q}_c a_c^{(k)} \;+\; \hat{q}_c^2 b_c^{(k)}, \qquad \bar{R}_c = \frac{1}{K}\sum_k R_c^{(k)}, \tag{9}$$

normalized per scenario (shifted to minimum zero, scaled by the median, clipped) so that all leak rates share one scale. Every image is closed-form linear algebra computed from the network's own inputs. Measured on an NVIDIA L40S at batch size one over 500 test scenarios after warm-up, host–device transfers included: preprocessing takes 29.6 ms median (31 ms p95) and the network forward pass 0.8 ms, for 30.4 ms median end to end. We verified (9) against direct reconstruction to relative error 10^{-7}.

Zero-budget reduction. With $\varepsilon = 0$ there is a single wind, so $v_c \equiv 0$, and the fit-quality image collapses onto the evidence image through the identity

$$R_c \;=\; \rho \;-\; \sigma^2 \max(z_c, 0)^2, \tag{10}$$

a fixed per-scenario transformation of z_c. The four images therefore carry exactly the information of the classical pair $(z_c, \log b_c)$: one recipe covers both settings, and the exact-wind results below are its zero-budget instantiation.

D. Training, the guarantee, and the division of labor

The input processor attaches to any base learner f_θ over the token set x through a three-layer convolutional head h_ϕ ($\sim$10k parameters, 0.2% of the model) that reads the map stack $M(x)$, joined at the output logits:

$$\ell(x) \;=\; f_\theta(x) + h_\phi\big(M(x)\big), \qquad \hat{p}(c \mid x) = \mathrm{softmax}\big(\ell(x)\big)_c, \tag{11}$$

with h_ϕ initialized at exactly zero, so at the first training step the model is the unmodified baseline and every comparison is controlled by construction. Training is ordinary supervised learning on smoothed one-hot targets $t(x)$:

$$\mathcal{L}(\theta, \phi) \;=\; -\sum_c t_c(x)\,\log \hat{p}(c \mid x). \tag{12}$$

Physics enters only through the inputs in (11).

For the guarantee we use split conformal prediction with a randomized highest-probability score. For a scenario with true cell c^*, the score is the probability mass ranked strictly above the true cell, plus a uniform share $u \sim \mathrm{Unif}[0,1]$ of the tied mass:

$$s(x, c^*) \;=\; \sum_{c:\,\hat{p}_c > \hat{p}_{c^*}} \hat{p}_c \;+\; u \sum_{c:\,\hat{p}_c = \hat{p}_{c^*}} \hat{p}_c. \tag{13}$$

The threshold $\hat{t}$ is the $\lceil (n{+}1)(1{-}\alpha) \rceil$-th smallest score over n calibration scenarios ($\alpha = 0.1$), and the region keeps the most probable cells, excluding only mass strictly below the threshold (the tie-handling correction of §III-A):

$$\mathcal{R}(x) = \big\{c : s(x, c) \le \hat{t}\,\big\}, \qquad \Pr\big[c^* \in \mathcal{R}(x)\big] \;\ge\; 1 - \alpha. \tag{14}$$

Calibration is always performed under the same wind condition in which the system is evaluated, so (14) certifies the deployment setting. Region sizes are reported as the radius of the circle of equal area; all models share identical data (10k/1k/2k/2k splits, a standard budget for this field [7]) and identical symmetry augmentation.

Scope of the guarantee. The conformal guarantee (14) is marginal over exchangeable scenarios drawn from the calibration distribution; it does not imply 90% coverage for every individual scenario, leak-rate range, sensor layout, or wind condition (subgroup diagnostics are summarized in Limitations). Calibration is performed separately for each evaluated deployment condition. The exact-model reference is used only to audit region efficiency under the synthetic testbed assumptions and is not part of the conformal guarantee.

Division of labor. The physics computes exact per-cell evidence through (6)–(9), knowledge a network would otherwise have to rediscover from data; the network, trained through (12), generalizes across layouts, rates, and wind histories and learns how much to trust each cell's evidence, knowledge with no closed form; conformal calibration (14) certifies the result.

Why not invert the physics directly? On this testbed one can; that is our exact-model reference (5). Deployed dispersion models have no exact solution, inversion takes minutes to hours per event [9], and §IV-C shows the evaluated inversion baselines are uninformative under measured wind.

TABLE I

THE PHYSICS IMAGES SHRINK GUARANTEED REGIONS FOR EVERY ARCHITECTURE (MEDIAN RADIUS IN M; EXACT WIND; 3 SEEDS PER CELL; COVERAGE 0.88–0.92 THROUGHOUT).

Base network	Without images	With images
DeepSets	94–98	**47.1–47.7**
GNN	56–57	**46.2–47.5**
Set Transformer	69–71	**46.2–47.3**

TABLE II

MEASURED WIND (10°/10% PER-STEP ANEMOMETER ERROR; NO METHOD OBSERVES THE TRUE WIND; CALIBRATION UNDER THE DEPLOYMENT CONDITION; DEEPSETS BASE; 3 SEEDS FOR LEARNED ROWS).

Pipeline	Cov.	Radius (m)	<50 m
Direct physics inversion	1.00	282 (full site)	0%
Network, no physics images	0.90–0.91	107.9–110.5	0%
Ours, uncertainty-aware images	0.90–0.92	**70.1–72.9**	**24–27%**

IV. RESULTS

A. The audit: a guarantee is not yet a useful region

Both learned pipelines pass every standard calibration check (coverage 0.89 at nominal 90%); only the audit reveals what calibration alone cannot. The labels-only baseline delivers 94–98 m where the exact-model reference is 6.2–7.0 m, more than ten times the reference; two input images and a small head close half of the distance, to 47.1–47.7 m. Relative to the labels-only model on the same 2,000 paired scenarios, the images reduce the median radius by 46.1 m (paired-bootstrap 95% CI $[43.6, 48.1]$), produce smaller regions in 96.4% of cases, and place 52.4% of regions below the 50 m facility scale (Clopper–Pearson 95% CI $[50.2, 54.6]$%); radius percentiles p50/p75/p90 are 48/86/131 m. The median case reaches the scale, and the upper tail marks the clearest room for future work.

B. The same fix works on every architecture

Without the maps, the choice of network changes results by 40 m; with them, all nine runs of three network families land inside a 1.5 m band (Table I). The gain is available to whatever architecture a practitioner already has, with ordinary training and near-zero added cost. At the other extreme, the classical score alone identifies where the evidence lies but not how much to trust it, and pays for that overconfidence with full-site regions. Single-seed channel diagnostics point the same way: the evidence image alone reaches 65.0 m, the sensitivity image alone 98.5 m, both channels 47 m.

C. Measured wind: the main result

Real anemometers have error, and the effect on direct physics inversion is dramatic. Trusting the measured wind exactly, it concentrates its probability so far from the true source that, to keep its guarantee, the calibration wrapper must return essentially the entire site (Table II); allowing extra measurement noise, tuned on held-out data, does not change

TABLE III
EACH IMAGE COMPONENT TIGHTENS THE REGIONS (DEEPSETS BASE, MEASURED WIND; 3 SEEDS; †: RANDOM-IMAGE NEGATIVE CONTROL, 1 SEED, DIAGNOSTIC).

Input images	Cov.	Radius (m)	$<$50 m
none	0.90–0.91	107.9–110.5	0%
random control†	0.89	107.5	0%
single-wind evidence pair	0.91–0.92	82.2–84.2	15–17%
+ moment images	0.89–0.91	74.7–76.3	20–21%
+ fit-quality image	0.90–0.92	**70.1–72.9**	**24–27%**

this. A negative control isolates the cause of our gains: feeding the identical head four *random* images of the same shape and scale (†, one seed, diagnostic) matches the no-physics baseline (107.5 m), so the improvement traces to the physics content of the images, not to added capacity or spatial regularization. The learned recipe behaves differently, and the pattern is clean: the more of the wind sensor's error-range information the input images carry, the smaller the guaranteed regions become (Table III): from 82–84 m using the measured wind alone, to 74–76 m after adding the moment images (8), to **70–73 m** after adding the fit-quality image (9). Relative to the no-physics model on the same 2,000 paired scenarios, the full method reduces the median radius by 39.4 m (paired-bootstrap 95% CI $[37.3, 41.9]$), produces smaller regions in 96.2% of cases, and places 26.6% of regions below 50 m (Clopper–Pearson 95% CI $[24.7, 28.6]$%). Throughout, this is one network, one objective (12), one guarantee (14), one forward pass. The mechanism is constructive: because the physics and the sensor's own error range live in the inputs, ordinary supervised training gives the network thousands of examples from which to learn how the error behaves and how much to trust each cell's evidence. In summary: $94 \to 47$ m from supplying a missing computation; $47 \to 71$ m, the measured cost of imperfect wind; $108 \to 71$ m, the value of physics in the inputs at this setting.

V. LIMITATIONS

CH4-T is a measuring instrument, not a deployment: single steady sources, flat terrain, known stability class, and a 64×64 grid whose $\sim$7.8 m cells sit near the reference scale. The 6–7 m reference is a property of the testbed's assumptions, exists only under known wind (the measured-wind analogue is not computable, so 70–73 m is reported against the 50 m operational scale), and the wind error range is taken as known from the instrument specification; quantifying sensitivity to a misspecified range is a natural next step. With exact wind, roughly half of individual cases still exceed 50 m (p90 = 131 m); prespecified subgroup diagnostics show no collapsing group, with mild under-coverage for sparse-mast scenarios as the one consistent pattern (coverage 0.865–0.867 at 4–6 masts, n=690, about 3 standard errors below nominal), a concrete target for group-conditional calibration. Field validation on controlled releases is the natural next stage, and the comparisons reported here are exact regardless, because every method shares one verified simulator, one dataset, and one calibration harness.

VI. CONCLUSIONS

On a testbed with exact ground truth, a calibrated deep localizer delivered a tenth of the sharpness of the exact-model reference. Supplying one classical computation as input images brought three network families to 46–48 m at the 50 m facility-association scale, and the same recipe, carrying the wind sensor's stated error range in its inputs, retained its guarantee at 70–73 m at the evaluated anemometer-error setting, at 30 ms per scenario with 0.2% added parameters.

The research impact is a transferable evidence standard: exactly solvable testbeds as measuring instruments for learned inference, self-checking calibration gates, and one-change-at-a-time controls, applicable wherever guaranteed regions are produced. The social impact is practical: regions small enough to act on, produced in about 30 ms from inexpensive hardware, with stated guarantees and the wind sensor's imperfection budgeted honestly. This provides a controlled step toward converting methane detections into calibrated inspection regions.

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Appendix

Appendix A: Full Results

All numbers are median 90% conformal region radii in meters over the 2,000-scenario test set, with empirical coverage and the fraction of regions below the 50 m facility scale; ranges are over 3 training seeds unless marked ^{1s} (single-seed diagnostic) or ^{2s}.

TABLE IV
Exact wind ($\varepsilon = 0$): all evaluated configurations.

Configuration	Estimator	Coverage	Radius (m)	<50 m
Exact-model reference	—	0.89–0.95	6.2–7.0	72%
Labels only	DeepSets	0.890	94–98	0%
Labels only	GNN	0.881–0.904	56.3–57.4	36–39%
Labels only	Set Transformer	0.888–0.894	68.7–70.8	18–22%
+ physics images	DeepSets	0.888–0.895	47.1–47.7	52–53%
+ physics images	GNN	0.891–0.902	46.2–47.5	53–54%
+ physics images	Set Transformer	0.892–0.901	46.2–47.3	53–54%
Flow-based estimator	spline flow	0.90–0.91	62.8–71.2	19–37%
Flow + physics images	spline flow	0.89–0.91	57.6–61.2	37–42%
Classical score alone	no network	1.00	282 (full site)	0%
Channel ablation: evidence only	DeepSets	0.88	65.0^{1s}	33%
Channel ablation: sensitivity only	DeepSets	0.90	98.5^{1s}	0%

TABLE V
Measured wind (10°/10% per-step anemometer error; no method observes the true wind; calibration under the deployment condition).

Configuration	Coverage	Radius (m)	<50 m
Direct physics inversion	1.00	282 (full site)	0%
Inversion, tuned noise inflation ($c^*\!=\!1.5$)	1.00	282 (full site)	0%
Network, no physics images	0.898–0.907	107.9–110.5	0%
Random images, identical head^{1s}	0.889	107.5	0%
Ours, single-wind images	0.911–0.920	82.2–84.2	15–17%
Ours, + moment images	0.894–0.906	74.7–76.3	20–21%
Ours, full (+ fit quality)	0.900–0.919	70.1–72.9	24–27%
Training-condition ablation:			
single-wind images, trained exact wind	0.908–0.912	84.1–85.1	21–22%
single-wind images, trained both conditions^{2s}	0.907–0.917	83.2–84.0	18%

TABLE VI
Rate-stratified performance of the recipe (exact wind, seed 1).

Leak rate (kg h^{-1})	n	Median radius (m)	<50 m
10–30	582	44.8	54%
30–100	585	44.5	55%
100–250	483	50.3	49%
250–500	350	55.2	41%

Reading the ablations. Both physics images contribute and the evidence image (the correlation computation) carries most of the gain; the sensitivity image alone, which never touches the readings, adds little over the baseline. The classical score without a network is overconfident-wrong and pays with full-site regions, so the learned calibration is load-bearing. The random-image control matches the no-physics baseline, so the improvement traces to the physics content of the images rather than to added parameters or spatial regularization. The training-condition ablation shows the measured-wind gains come mainly from the images themselves rather than from training under the deployment condition.

Appendix B: Paired Bootstrap Details

For each comparison, both methods are evaluated on the same 2,000 test scenarios; we resample scenarios jointly (10,000 bootstrap draws), so every draw contains paired results, and report the difference of median radii with percentile 95% intervals. Fractions below 50 m carry exact Clopper–Pearson 95% intervals. Exact wind, labels-only vs. recipe: median reduction 46.1 m,

TABLE VII

SUBGROUP DIAGNOSTICS (PRESPECIFIED GROUPS; SEED-1 RUNS). THESE ARE DIAGNOSTIC; THE FORMAL CONFORMAL GUARANTEE REMAINS MARGINAL OVER THE COMPLETE TEST DISTRIBUTION.

		Exact (ours)		Measured (full)	
Subgroup	n	Cov.	Med.	Cov.	Med.
4–6 masts	690	0.865	61.2	0.867	91.2
10–12 masts	675	0.899	42.7	0.926	59.5
10–100 kg/h	1167	0.881	43.2	0.913	80.6
100–500 kg/h	833	0.904	53.8	0.881	61.6
wind $<$ 2.5 m/s	1000	0.885	49.6	0.884	63.0
wind $\geq$ 2.5 m/s	1000	0.896	46.8	0.916	79.8

TABLE VIII

REPRODUCIBILITY SUMMARY; FULL CONFIGURATION AND CODE IN THE PROJECT REPOSITORY.

Setting	Value	Setting	Value
Split	10k/1k/2k/2k	Optimizer	AdamW, lr $3\times10^{-4}\rightarrow3\times10^{-5}$ cosine, wd 10^{-4}
Site, grid	500 m; 64^2 (7.8 m cells)	Batch, epochs	256; 200, best validation-NLL state
Masts	4–12, count/positions uniform	Precision	bfloat16 autocast, grad clip 1.0
Readings	30 $\times$ 1-min averages; dropout $\leq$20%	Targets	one-hot, Gaussian-smoothed σ=0.75 cells; D4
Noise σ	log-uniform 0.2–3 ppm, iid	z, $\log b$ normalization	clip $\pm$60, $\div$10; $(\log_{10} b+8)/6$
Leak rate	log-uniform 10–500 kg h^{-1}	Residual normalization	min-shift, median-scale, clip 10
Wind	speed log-unif. 1–8 m s^{-1}; AR(1) ρ=0.7	Parameters	4.88 M total; head 10,721 (0.22%)
Meander	20–40° dir., 10–30% speed	Seeds	fixed data root; training 1–3; fixed score seed
Stability	PG classes A–F, uniform	Hardware	NVIDIA L40S
Wind error	10° dir., 10% speed, per step	Timing (batch 1)	maps 29.6 ms, net 0.8 ms; 30.4 med, 31.7 p95

CI $[43.6, 48.1]$; 96.4% of scenarios improved; 52.4% below 50 m, CI $[50.2, 54.6]$%. Measured wind, no-physics vs. full method: median reduction 39.4 m, CI $[37.3, 41.9]$; 96.2% improved; 26.6% below 50 m, CI $[24.7, 28.6]$%.

APPENDIX C: ADDITIONAL RESULT FIGURES

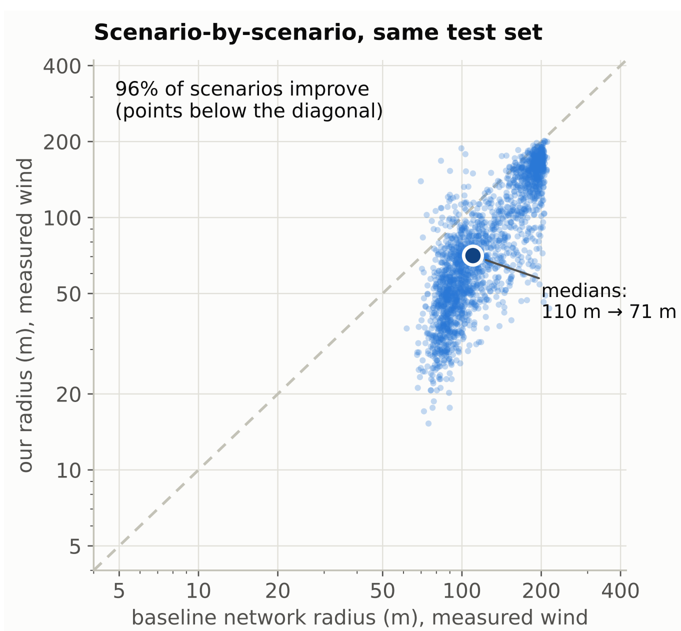

Fig. 3. Scenario-by-scenario paired comparison under measured wind (same 2,000 test scenarios): 96% of points lie below the identity line; medians 110 m $\rightarrow$ 71 m.

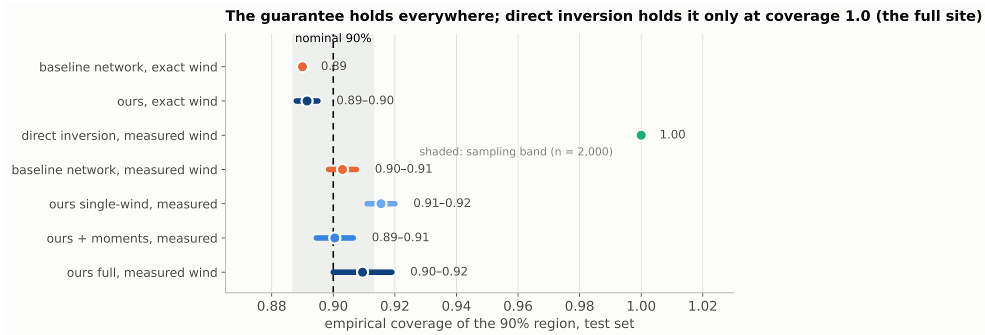

Fig. 4. The guarantee check: empirical coverage of every configuration against the nominal 90% line with the n=2,000 sampling band. Direct inversion holds its guarantee only at coverage 1.0, i.e., by returning the full site.

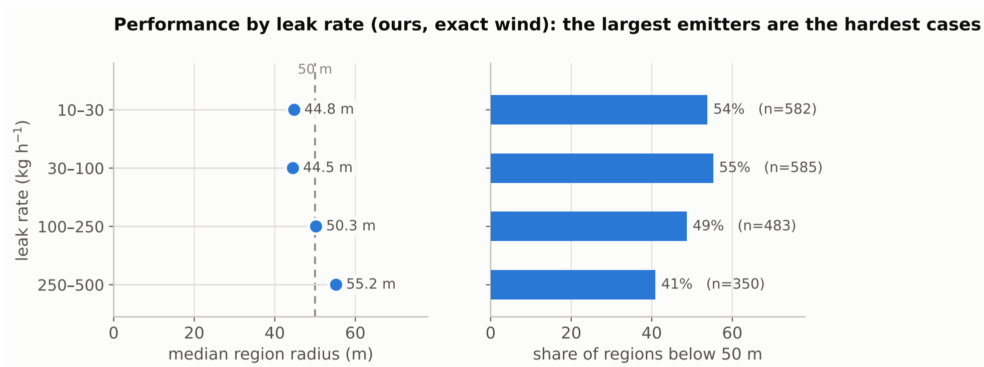

Fig. 5. Performance by leak rate (ours, exact wind): the largest emitters are the hardest cases.

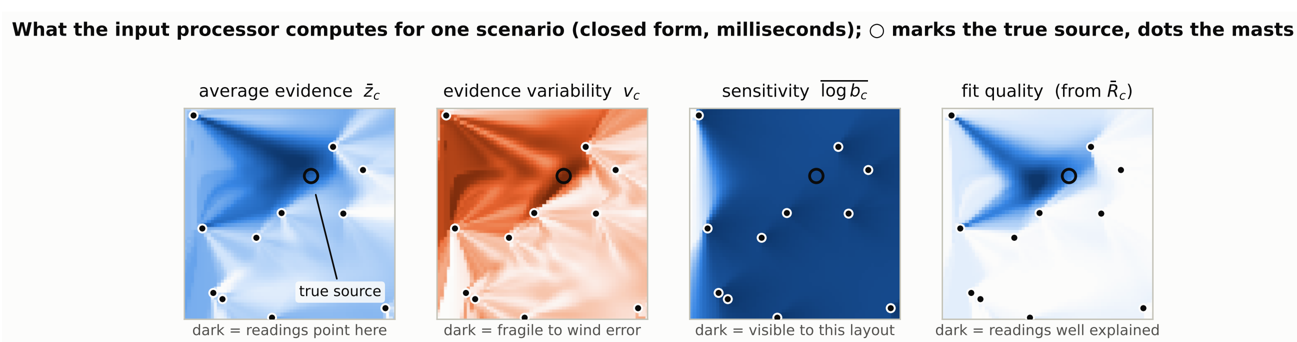

Fig. 6. The four physics input images for one test scenario (closed form, computed from the network's own inputs at the measured wind): average evidence, evidence variability across the wind-error range, sensitivity, and fit quality. The circle marks the true source; dots mark the sensor masts.